



Acid Rain Detectives



How can we detect acid, and what does acid rain do to stone?

Smoke from factories and cars can make rain a little bit acidic. Acid rain can harm plants and wear away stone. Let's become acid detectives!

TIME

60 minutes

GROUPS

Groups of 3, with an adult nearby

AGES

Starter 6–8 · Explorer 9–11

Learning goals

- ✓ I can use a colour tester to find an acid.
- ✓ I can sort liquids by the colour they make.
- ✓ I can explain how acid rain damages stone.

STEAM in this activity

S

SCIENCE

Acids and bases; a reaction makes gas bubbles.

T

TECHNOLOGY

An indicator is a detection tool.

E

ENGINEERING

Think of ways to protect a stone statue.

A

ART

Paint with cabbage water and make colours change.

M

MATHS

Sort results into a table; Explorers count bubbles.



Before the lesson

- ☐ Make the cabbage water the day before (adult). Keep it cool.
- ☐ Test your limestone chips or eggshell in vinegar. Classroom chalk often does not fizz.
- ☐ Safety glasses, labelled clear cups, spoons and paper towels.
- ☐ Water, vinegar and baking soda water ready in jugs.



Safety

- Only an adult uses the knife and hot water.
- Wear safety glasses.
- Never drink or taste anything in science.
- Wash your hands when you finish.
- Tell an adult if anything splashes in your eyes.



Kit per group

- ☐ 🥬 A few red cabbage leaves
- ☐ 🍵 Hot water (an adult pours it)
- ☐ 🍲 A bowl, a sieve and a jug
- ☐ 🍷 Three clear cups
- ☐ 🧴 Water, vinegar and baking soda water
- ☐ 🪨 Two limestone chips or eggshell pieces
- ☐ 🍷 Two clear cups
- ☐ 🍷 Water and vinegar

Running order (60 minutes)

0–10 min	Story: where does acid rain come from?
10–20 min	Build: show how the tester was made (or make it).
20–35 min	Test 1: colour detectives.
35–50 min	Test 2: stone in water and in vinegar.
50–60 min	Share, clear up and wash hands.

The investigations

Start here: build: Make your colour tester

Children record: What colour did it turn?; Draw your tester

What scientists often see: Red cabbage has a colour inside it that changes when it meets an acid or its opposite, a base. Scientists call a colour tester an indicator.

Test 1: Colour detectives: Which liquid will turn the tester pink?

Children record: What colour did it turn?

What scientists often see: Acids turn red cabbage pink or red. Plain water keeps it purple. Bases, like baking soda, turn it blue or green. Vinegar is a weak acid.

Test 2: Acid on stone: What happens to stone in vinegar?

Children record: How many bubbles?; Draw what you see

What scientists often see: Limestone and eggshell are made of a stuff that acids break down. The bubbles are a gas made as the stone wears away. Acid rain slowly wears away stone statues and buildings the same way.

Talk about it

- How do we know vinegar is an acid?
- What made the bubbles on the stone?
- How could we protect statues from acid rain?

What to look for

Starter: Matches pink to acid and circles the right colours.

Explorer: Explains that acid reacts with stone and makes a gas.

Art corner

Magic colour painting. Paint a picture with cabbage water. Dab on vinegar for pink and baking soda water for green. Stick or draw it here.

Maths corner

Starter: Colour in the cup for each liquid, the colour it turned.

Explorer: Count the bubbles you see in 10 seconds in each cup: water ___, vinegar ___.

Real-world link

Old stone statues and buildings in cities slowly wear away in acid rain. Cleaner air helps protect them.

Going further

Test more liquids: lemon juice, soapy water, milk. Acid or not?

